David Henderson, Ph.D.

+1 (312) 768-0111 | dhenders013@proton.me
San Francisco Bay Area
LinkedIn **|** GitHub **|** Google Scholar

SUMMARY

Ph.D. in Ecology, Evolution, and Population Biology with 6 years of experience in genomics, bioinformatics, and advanced data analytics. Demonstrated expertise in analyzing high-dimensional and multi-omics datasets, integrating machine learning and statistical methods to uncover biological insights. Proficient in R, Python, and cloud-based analytical tools (AWS). Track record of peer-reviewed publications in novel omics analyses and ecological modeling, with a passion for leveraging computational tools to advance research in biodiversity, evolutionary biology, and ecosystem genomics.

RESEARCH EXPERIENCE

Ph.D. Research Biologist **2018—2023**

Department of Biology, Washington University in St. Louis

- Led multi-omics research projects integrating proteomic, metabolomic, and environmental datasets to assess the impact of pathogens and natural enemies on tropical forest trees across elevation gradients.
- Developed and implemented bioinformatic pipelines for processing high-dimensional and mass spectrometry data, employing R/Bioconductor and Python for variant analysis, expression profiling, and network analysis.
- Characterized chemical defense mechanisms by combining proteomic and metabolomic data, utilizing machine learning and statistical modeling to identify key functional and metabolic markers associated with species distribution.
- Co-organized and led an Andes-Biodiversity working group of 20+ people, in English and Spanish, with collaborators from South America, Europe, and the United States, which set the foundation for the above projects.

Visiting Faculty **2022**

Dr. Brian Sedio Lab, University of Texas at Austin

- Led the development of scalable bioinformatics pipelines for large-scale metabolomic and genomic data analysis, including sample processing, quality control, and data integration.
- Applied advanced analytical techniques (e.g., differential abundance analysis, pathway enrichment) to multi-omics datasets, collaborating with computational biologists and software engineers to optimize workflows for high-throughput data.

TECHNICAL SKILLS

Genomics & Bioinformatics: Next-generation sequencing (NGS), Lab Preparation Protocols (DNA, RNA, CRISPR, ATAC-Seq), genome assembly, variant calling, gene expression analysis, annotation, Bioconductor, Biopython, Population Genetics (FastQC, Trimmomatic, BWA, SAMtools, GATK, FST)

Data Analysis & Modeling: R, Python (pandas, scikit-learn, Biopython), SQL, Linux, AWS Cloud for analytics, machine learning, multivariate analytics, Metabolomics and mass spectrometry tools (Mzmine2, GNPS, Qemistree, CSI:FingerID, ClassyFire, Interactive Tree of Life (iTOL) tool), Population Genetic Analyses

Visualization: Tableau, ggplot2, Seaborn, Shiny, Streamlit, Flask, FastAPI

Other: GIS, Earth-system modeling, SDM

EDUCATION

Ph.D. in Ecology, Evolution, and Population Biology
Division of Biology and Biomedical Sciences, Washington University in St. Louis, MO, USA
2018–2023

> **Dissertation:** *Dissertation: Biotic Interactions and the Maintenance of Biodiversity Across Biogeographic Gradients*
> **Advisor:** Dr. Jonathan Myers
> **Committee:** Dr. J. Sebastian Tello, Dr. Rachel Penczykowski (Chair), Dr. Brian E. Sedio, Dr. Adam B. Smith

B.Sc. in Biological Sciences, *summa cum laude*
Biology Department, Northern Illinois University, DeKalb, IL, USA
2014–2016

RECENT EMPLOYMENT

Software Developer **2024**

Galactic Advisors

- Utilized strict version control and development processes, contributing to the daily operations of the quality control team.
- Maintained and created multi-structured database features. Optimized query rules for data transfer, analyses and automatic processes including data upload, preprocessing, formatting, analysis, and report generation.
- Updated most critical, most used sections of database schema within the first 30 days.
- Worked across security research team, software development team, and compliance teams to produce deliverables within a short and urgent timeframe.
- Technical tools and proficiencies: CI/CD, Git, SQL, Python, Linux, Laravel, and encryption management tools.

Research Program Mentor **2023 – Present**

Polygence

- Worked with more than 15 dedicated high school students through completion of self-directed research projects where they learn how to be a scientist, including the process of rigorous scientific inquiry via experimental design, scientific writing and data analysis.
- Maintain active communications with students and parents to ensure consistent progress throughout the projects.
- Simultaneously mentor 5 student research projects in ecology and environmental science, aiming at co-authoring publications.

Biotech Consultant **2021—2023**

The Biotechnology and Life Science Advising (BALSA) Group

- Worked as a project-based consultant and team member of a non-profit, student-led consulting group.
- Selected appropriate statistical tools to address specific client needs. Summarized findings in written and oral forms.
- Performed primary and secondary research including Surveys and Interviews with KPIs, Market Segmentation, Market Sizing, Competitive Analyses (SWOT), and Exit Strategies research for academic inventors, biotech innovators and investors, in a team-based environment.
- Stayed abreast of the latest developments, innovations, regulations, and standards relevant to the project.

PEER REVIEWED PUBLICATIONS

***Published*:**

4. Henderson, D., Sedio, B. E., Tello, J. S., Cayola, L., Fuentes, A. F., Alvestegui, B., Muchhala, N., & Myers, J. A. (2025). Testing the role of biotic interactions in shaping elevational diversity gradients: An ecological metabolomics approach. Ecology, 106(4), e70069.

3. Smith, A. B., Murphy, S. J., Henderson, D., & Erickson, K. D. (2023). Including imprecisely georeferenced specimens improves accuracy of species distribution models and estimates of niche breadth. Global Ecology and Biogeography, 32(3), 342-355.

2. Rader, A. M., Cottrell, A., Kudla, A., Lum, T., Henderson, D., Karandikar, H., & Letcher, S. G. (2020). Tree functional traits as predictors of microburst-associated treefalls in tropical wet forests. Biotropica, 52(3), 410-414.

1. Hazzi, N. A., Petrosky, A., Karandikar, H., Henderson, D., & Jiménez-Conejo, N. (2020). Effect of forest succession and microenvironmental variables on the abundance of two wandering spider species (Araneae: Ctenidae) in a montane tropical forest. The Journal of Arachnology, 48(2), 140-145.

In Progress:

2. Chadwick, S.E., Henderson, D, Cayola, L., Fuentes, A. F., Muchhala, N., Tello, J. S., Volf, M., Myers, J. A., Sedio, B. E. (2025). Chemical properties of foliar metabolomes represent a key axis of functional trait variation in forests of the tropical Andes. (Submitted to PNAS).

1. Henderson, D., Sedio, B. E., Tello, J. S., Cayola, L., Fuentes, A. F., Loza, M. I., Wu, Y, & Myers, J. A. (2024). Disentangling determinism and stochasticity in local tree neighborhoods across a tropical elevational gradient. (in prep for Global Ecology and Biogeography).

AWARDS AND GRANTS

- 2022: Botanical Society of America Travel Grant $375
- 2022: American Society of Plant Taxonomists Graduate Student Grant $500
- 2022: Washington University Division of Biology and Biomedical Science Graduate Student Travel Grant $600

- 2022: Maxwell Hanrahan Foundation-Missouri Botanical Garden Funds (Co-PI), $3,000
- 2021: Maxwell Hanrahan Foundation-Missouri Botanical Garden Funds (Co-PI), $3,400
- 2020: ESA SEEDS Interdisciplinary Power of Data Research Travel Award Ecological Society of America $500
- 2020: Living Earth Collaborative for “Testing the Role that Biotic Interactions Play in Shaping Elevational-Diversity Gradients: An Ecological Metabolomics Approach” (co-author), $29,010
- 2020 – 2023: Catharine M. Lieneman Scholarship in Botany Washington University in St. Louis for full stipend amount

TEACHING EXPERIENCE

- 2022: Nominee Washington University in St. Louis Graduate Teaching Award
- 2021: Washington University Center for Teaching and Learning - Professional Development in Teaching Program completed – Certificate awarded

Teaching:

- 2020: Teaching Assistant, Community Ecology
- 2019: Teaching Assistant, Natural History of Missouri

Guest Lecturer:

- 2019–2023: Woody Plants of Missouri Systematics (undergraduate level), Washington University of St. Louis

Other Teaching Experience:

- 2022: Teaching Assistant, Vertical Voyage Tree-Climbing Class for K-12 Students, St. Louis, U.S.A

PROFESSIONAL DEVELOPMENT

- Google Analytics Academy: Google Analytics Certification (2025)
- Amazon Web Services: Certified Machine Learning – Specialty certification (2025)
- Amazon Web Services: Certified AI Practitioner certification (2024)
- Course Participant: Sample to Genome Course, Wellcome Sanger Institute, Hinxton, Cambridgeshire, UK (2024)
- 2021-2022: Bioinformatics and Lab Sample Training with Sedio Lab, University of Texas at Austin, Virtual - Austin, TX, U.S.A.

- 2022: Geographic Information Systems Applications Course – Environmental Studies 380, Washington University, St. Louis, MO, U.S.A.
- 2021: Power of Data Workshop, Ecological Society of America, Baltimore, MD, U.S.A.
- 2021: Tree Climbing Course, Vertical Voyages, St. Louis, MO, U.S.A.
- Washington University Wilderness First Aid practical course: Instructor Dr. Stanton Braude
- Professional Association of Diving Instructors (PADI) certified Rescue Diver

PRESENTATIONS

- 2022: Nov Biology Lab Talk – Metabolomic pipeline overview (for ecologists)
- 2022: Botanical Society of America Conference: Testing the role that biotic interactions play in shaping elevational-diversity gradients: An ecological metabolomics approach. Botanical Society of America Conference, Anchorage, AK.
- 2022: Jul Biology Lab Talk – Ecological Scale Metabolomics
- 2022: May BioForum – "Testing the Role that Biotic Interactions Play in Shaping Elevational Diversity Gradients: An Ecological Metabolomics Approach".
- 2021: Canopy Science Forum – "Lonely Oaks, Quercus humboldtii", virtual.
- 2020: Biology Department, University of Missouri - St. Louis, St. Louis, MO
- 2019: Ecological Society of America Conference: "Ecological Data Modeling for Asclepias". Louisville, KY.
- April 2019 Midwest Ecology and Evolution Conference: "Ecological Data Modeling for Asclepias".
- Terra Haute, IN.
- 2019: "Sapotaceae" for Angiosperm Phylogeny Plant Systematics group
- 2018: "Boraginales" for Angiosperm Phylogeny Plant Systematics group
- 2018: Danforth Microbe Meetup "Urban transects: Invaders as a prediction of infection".

Selected Washington University BALSA Group seminars and workshops

- 2021: BALSA Summer Professional Development Series Session 1: How Research Gets from the Bench to the Patient Bedside, BALSA Group, St. Louis, MO, U.S.A.
- 2021: BALSA Summer Professional Development Series Session 2: Running Inclusive Meetings, BALSA Group, St. Louis, MO, U.S.A.
- 2021: BALSA Summer Professional Development Series Session 3: Business Communication, BALSA Group, St. Louis, MO, U.S.A.
- 2021: BALSA Summer Professional Development Series Session 4: Effective Slide Design for Business Meetings, BALSA Group, St. Louis, MO, U.S.A.
- 2020: Botany 2020 Conference Workshop, "Ace it! - Give a Better Talk"

Selected Washington University Center for Teaching and Learning seminars and workshops

- 2021: Creating a Teaching Portfolio, Developing Effective Summative Assessments

- 2020: Writing a Teaching Philosophy Statement, Engaging in Equity Pedagogy in the STEM Classroom, Crickets in the Classroom: Quick tips for Jumpstarting Conversation when Participation Lags
- 2019: Fostering an Inclusive Classroom Climate, Who's in Charge? Negotiating Your Role and
- Establishing Authority in the Classroom, Constructing Effective Collaborative Learning
- Opportunities, Using the Jigsaw Method to Increase Student Engagement
- 2019: Andes Working Group - Washington University, Tyson Research Center, Eureka, MO.
- 2019: Wilderness First Aid course – Washington University, Tyson Research Center, Eureka, MO.
- 2019, R Summer Workshop, Missouri Botanical Garden, St. Louis, MO, U.S.A
- 2018: Introduction to Pedagogical Scholarship Workshop, Planning and Organizing a Class Session, Facilitating Challenging Conversations, Motivating Student Learners
- 2018, Tropical Field course of Organization of Tropical Studies, Topic "Tropical Biology: An Ecological Approach," Costa Rica

SERVICE AND OUTREACH

- 2024 – Present: Washington University Bay Area Young Alumni Network, Co-Chair
- 2024: Volunteer CaveSim. San Mateo, CA
- 2023: Plant Sale: Harris World Ecology Center, University of Missouri in St. Louis
- 2023: Volunteer Festival of Nations, Missouri Botanical Garden, St. Louis, MO
- 2022: Plant Sale: Harris World Ecology Center, University of Missouri in St. Louis
- 2022: Panel Member - Washington University in St. Louis, Association of Black Biomedical Graduate Students
- 2022: Volunteer - Vertical Voyages, Columbia, MO
- 2021: Plant Sale: Harris World Ecology Center, University of Missouri in St. Louis
- 2020: Participant - Washington University in St. Louis
- 2020: Mentor - to incoming graduate students: Washington University in St. Louis, Initiative to Maximize Student Diversity
- 2019: Mentor - to incoming graduate students: Washington University in St. Louis, Initiative to Maximize Student Diversity

Referee*:*

- Biotropica, 2021
- Oecologia, 2020

SOCIETY AFFILIATIONS

- Washington University Young Alumni Network: Bay Area Co-Chair
- Ecological Society of America
- American Society of Plant Taxonomists
- American Society of Naturalists
- Botanical Society of America
- Association of Black Biomedical Graduate Students
- Washington University Student Union (WUGWU)
- Washington University Initiative to Maximize Student Diversity

LANGUAGES

- English (fluent)
- Spanish (CEFR B2)

References:

- Dr. Jonathan Myers (jamyers@wustl.edu)
- Dr. Yingtong “Amanda” Wu (ytwu@stanford.edu)
- Dr. James Lucas (fiberethnobotanist@gmail.com)